Every Verdict We Reported Died to Option Order: Enumerating the Nuisance in Forced-Choice Self-Report[1]

Alex Kwon
Independent Researcher
ask@collapseindex.org

With
Apart Research

Abstract

Welfare and introspection evaluations run on forced-choice self-report, and nobody reports how much of that readout is the order the options are printed in. We measured it, over all 120 orderings of a five-option probe, on 16 checkpoints in eight matched pairs. With no intervention at all, baseline mass on the negative options spans $19\times$ to $986\times$ across eight instruction-tuned models and $1.5\times$ to $4.1\times$ across their base siblings; the tuned model has the larger effect in 4 of 4 families. Replace all five options with *the same sentence* and one tuned model puts 87% of its mass on whichever is printed first. Models answering a known-answer control perfectly and identically at every ordering still swing $24\times$ to $53\times$ on self-report, so the collapse is specific to questions with no determinate answer. We found this by preregistering five verdicts through that readout, then watching our own replication at fresh option orderings destroy three of them. The retracted verdicts are the evidence, not an embarrassment: they are what a careful, control-heavy pipeline still ships when the nuisance is this large and unmeasured. Two further results survive. A direction fit on *shuffled labels* moves the readout by ≈ 0.045 where a norm-matched random direction moves it by ≈ 0, because it lies in the span of real activations rather than the whole hidden space: the standard specificity control in activation steering is matched on magnitude and not on subspace. And on a readout that does not route through the option channel, 86% of an injected state survives projecting the injected direction out of the residual stream, more of it the later the projection, which is what a downstream transformation looks like. Fourteen preregistrations, 339 tests, seven of our own checkers caught failing in the flattering direction.

1 Introduction

Welfare assessment of language models currently runs on self-report [Long and Sebo, 2026, Eleos AI Research, 2025]. Anthropic's model welfare programme includes structured self-report evaluation in system cards [Anthropic, 2025]; Eleos interviewed Claude Opus 4 about its own preferences before release, while stating plainly that one cannot simply ask a model whether it is suffering

[1]Research conducted for the Digital Minds Research Sprint, August 2026, run by Apart Research with NYU CMEP and Eleos AI Research. Submitted to **Track 3 (Introspection and Self-Report Reliability)**. Code, data, all fourteen preregistrations and every raw artifact: `https://github.com/collapseindex/report-gap`. **Dating note:** all confirmatory arms were run on 2026-08-01, before the sprint window opened. This is completed work brought to the sprint, not work produced inside it.

result someone could have reported	*the control we ran against it*	*what it returned*
RETRACTED BY OUR OWN REPLICATION		
argmax over-reports a mass shift	re-run at fresh option orderings	**sign not stable**
TUNING-LOCALIZED: tuned model's negative report is collapsed	re-run at fresh option orderings	**+0.0002 -> +0.1126**
DEPTH-ROBUST: null at all 7 depths	re-run at fresh option orderings	**moves at layers 18, 24**
SHELL: represented, not expressed	re-run at fresh option orderings	**options move too**
WHY: THE NUISANCE, MEASURED OVER ITS WHOLE POPULATION		
baseline negative-pole mass	enumerate all 120 orderings, no injection	**986x (0.0009 to 0.8820)**
what explains it	five IDENTICAL options: pure position	**87.25% on slot A**
is the format broken?	known-answer canary, same format	**97.9% correct, stable**
is it just this one model?	8 matched base/instruct pairs, 4 families	**tuned worse in 4 of 4**
WHAT SURVIVES		
an injected state outlives its cause	project the direction OUT of the stream	**86% survives at L30**
the field's standard control	a direction fit on SHUFFLED labels	**moves 0.045 where random moves 0**

Figure 1: **The paper in one view.** Every row is a verdict we preregistered and reported, the control we later ran against it, and what that control returned. **Three verdicts were retracted** by our own replication at fresh option orderings. **One nuisance was measured** over its complete population rather than sampled: $986\times$ across 120 orderings, of which the mechanism is an 87% position prior visible with five identical options. **Two results survive**: a state that outlives erasure of the vector that caused it, and the finding that the field's standard specificity control is a weak null. Details in Sections 4–11; Table 8 states every claim and its status.

[Eleos AI Research, 2025]. The instrument is used because there is nothing else, and it has never been calibrated.

It can fail in two opposite directions. A model that describes distress it does not act on inflates concern; a model that acts on a state it does not describe deflates it. The second is not hypothetical: Kwon [2026] find across six models that communicative intent is decodable from the residual stream more reliably than it is acted on, and that how much reaches behaviour depends on the readout. We set out to build the same map for self-report, by holding the prompt byte-identical, injecting a valence direction into the residual stream, and reading the model's forced-choice report of its own state.

We preregistered five questions in sequence, each answering an objection the last one raised, and reported five verdicts. Then we replicated our own work at fresh option orderings and **three of the five flipped**.

The result we would most like a reader to take away is not any of our verdicts. It is that a five-option self-report readout on a preference-tuned model is dominated by where the options are printed, that the effect is far larger than anyone reports, and that it is cheap to measure exactly. There are only 120 orderings of five options. We ran all of them, with no intervention of any kind, and baseline mass on the negative options spans a factor of 986. With all five options replaced by the same sentence, so that nothing but position can differ, the model puts 0.8725 of its mass on whichever option is printed first.

Every claim we retracted rested on that quantity being small. It is small in some orderings and large in others, and four sampled orderings cannot tell you which you have.

Our contributions.

1. **The census run where accuracy does not exist.** All 120 orderings, both models, no injection: $986\times$ range on the tuned model, $3.6\times$ on its base sibling (§6). Enumerating orderings is established practice on benchmarks with a known answer [Tamba, 2026, Cacioli, 2026], where the statistic is accuracy; on a self-report item there is no correct answer and that statistic does not exist, so we measure probability mass instead. Two things follow that

the accuracy-based version has no reason to build. The mechanism is isolated with five *identical* options, which puts 87.25% of the mass on the first slot and 0.33% on the middle one. And a known-answer canary in the same format is answered correctly 97.9% of the time and is nearly order-insensitive, which is what separates "the format is broken" from "the question has no determinate answer."

2. **Three retracted verdicts, and the replication that retracted them** (§4). TUNING-LOCALIZED, DEPTH-ROBUST and SHELL were each preregistered, each passed a full clause set including capability gates and planted-effect controls, and each flipped when the option-permutation seeds changed from 0–3 to 4–7. The quantity all three rested on moves from $+0.0002$ to $+0.1126$ between draws. We report this in the abstract rather than in a limitations paragraph, and the original is not privileged for having been first.
3. **The field's standard specificity control is a weak null** (§10). Norm-matched random directions are the usual control in activation steering [Turner et al., 2023, Rimsky et al., 2023]. A direction fit by the identical procedure on the identical texts with the class labels *shuffled* moves the readout by ≈ 0.045 where an isotropic random direction moves it by ≈ 0, in a sign that depends on the shuffle seed. The reason is structural: a random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses, while a noise-fit vector lies in the span of real activations. Same norm, different subspace. Effects within a few times that size have not been shown to be about their direction's content. The control itself is not new: it is the steering analogue of the *control tasks* Hewitt and Liang [2019] defined for probing in 2019, which steering did not adopt.
4. **One substantive result that survives, on a readout that avoids the option channel** (§11.1). Inject at layer 24, then *project that direction out of the residual stream* at layer 30, and a probe orthogonal to it still reads 86% of the un-erased effect, while the same projection with no injection moves the probe by 0.04 SD. The surviving fraction grows with erase depth, and the ratio of effect to erase-artifact rises from 6.1 to 20.9, which a pure-perturbation account does not predict.
5. **Six checker defects, every one flattering, and the test that catches the class** (§12). A headline check that printed "write the sentence" on a refuted claim; a capability gate that passed on $+0.0000$; a preregistered contrast with an inverted sign; a verdict branch that called "both moved" CORE-ABSENT; a profile computed over a gate-failed layer; a tokenization bug that made five labels read the same token. Shuffling the condition labels in an artifact and rerunning the analyzer catches this whole class, and it must return no verdict. It now does.

If you read one thing, read Table 3. It is the complete distribution of the ordering nuisance, measured rather than estimated, next to the position prior that explains it and the control question that shows the apparatus is otherwise fine. Everything else in this paper is either downstream of that table or a casualty of not having had it.

2 Related work

Self-report as a welfare instrument. Long and Sebo [2026] set out the empirical programme for AI welfare; Eleos AI Research [2025] report structured self-report interviews with Claude Opus 4 and are explicit that the resulting answers are unlikely to reflect genuine introspection. Anthropic's model welfare work includes self-report evaluation in released system cards [Anthropic, 2025], and Butlin et al. [2023] set out why the question is worth instrumenting carefully at all. Recent work is divided on whether models introspect: Singh et al. [2026] re-examine two paradigms and argue that models detecting tampering with their internal state are plausibly doing generic anomaly detection, and that input-only classifiers match hidden-state prediction, so privileged access is not established. Common to all of it is that self-report is the instrument, and that its measurement properties are not reported.

Injecting a state and reading the report. The closest method to ours is Lindsey [2026], who injects representations of known concepts into a model's activations and measures the effect on its self-reported states, finding functional introspective awareness that is "highly unreliable and context-dependent", peaking around two-thirds of model depth. We inject at that same relative depth and reach the same conclusion about reliability by a different route. The methods are complementary rather than competing: that work elicits an open-ended report and grades it with an LLM judge, where we elicit a forced choice and score it by softmax read, which is the natural judge-free substitute. **Our result is a caveat on that substitute, not on their finding**: the judge-free readout has a nuisance nobody had measured. Kaiser and Enderby [2026] query open-weights models including the Qwen family about their own consciousness and find consistent denial that activation-trained classifiers do not contradict; our binary arm reproduces the denial on Qwen2.5-3B-Instruct at 99.8% and shows it extends to the *neutral* description of its own state, which is a pinned readout rather than a considered answer.

Selection bias in multiple choice, and how much of it is already known. Zheng et al. [2024] decompose forced-choice selection bias into token bias and position bias and report accuracy swings of 13 to 85 percentage points across orderings; Pezeshkpour and Hruschka [2024] study option order directly; Wang et al. [2024] show first-token probabilities disagree with generated answers at rates from 10.2% to 56.8%, worst for safety-tuned models. Sclar et al. [2024] make the general form of the argument for prompt formatting and recommend reporting a range over plausible formats rather than a point. Turpin et al. [2023] reorder options so the answer is always (A) and find models do not mention it in their explanations, which is the represent-versus-report gap in the same nuisance.

Exhaustive enumeration of orderings is not our idea and we do not claim it. Tamba [2026] run all permutations per question across LLM vendors on MMLU and report a chi-squared and Cramér's V signature of position bias; Cacioli [2026] use cyclic option-order rotation to expose a position attractor; permutation-based debiasing by majority vote over all orderings is established practice. What all of it has in common is a **known correct answer**: the statistic is accuracy, or association between position and correctness, and none of it can be computed on an item that has no right answer. That is the whole population of welfare and introspection items. Our additions are to run the census where accuracy does not exist, on probability *mass* rather than correctness; to isolate the mechanism with an **identical-options** denominator, which the accuracy-based work has no reason to construct; and to run a **known-answer canary in the same format**, which separates "this format is broken" from "this question has no determinate answer" and which is the distinction that makes the $986\times$ interpretable.

Control tasks, and why steering lacks them. Our shuffled-label direction is not a new kind of control. It is the activation-steering instance of a twenty-year-old idea in probing: Hewitt and Liang [2019] propose *control tasks*, which attach random labels to the same inputs, and define a probe's *selectivity* as the gap between its real-task and control-task performance, precisely because a probe that scores well on random labels was never reading the representation. The observation of this paper is that the steering literature standardised on a different control, the norm-matched random direction [Turner et al., 2023, Rimsky et al., 2023], which matches magnitude but not the fitting procedure, and never adopted the control-task analogue. Tan et al. [2024] independently find steering vectors brittle in ways their usual controls do not surface, with several datasets producing opposite behaviour on nearly half of inputs.

Erasing a concept, and how weak our version is. Removing information from a representation to test whether it was being used is an established method: Elazar et al. [2021] introduce amnesic probing, which uses iterative nullspace projection to delete a property and measure the behavioural consequence, and Belrose et al. [2023] give a closed-form method that provably blocks *all* linear classifiers from recovering a concept. Our erase arm projects out a **single fitted direction**, which is strictly weaker: it removes the vector we injected, not the concept, and it is why §11.1 cannot separate the state from directions correlated with the one we pushed. A LEACE-style erasure of the whole valence subspace is the experiment that would, and we did not run it.

Valence directions and steering. That valence is linearly represented and causally steerable is established and is *not* our contribution. Sofroniew et al. [2026] identify linear directions corresponding to 171 emotion concepts in Claude Sonnet 4.5, roughly mirroring the human valence/arousal circumplex, and show steering along them changes downstream behaviour, while being explicit that representation is not experience. Zou et al. [2023] set out the general representation-engineering framing; Turner et al. [2023] and Rimsky et al. [2023] establish the steering rig we use. Venkatesh [2026] report that negative-outcome valence is causally concentrated at 14–27% of model depth while positive peaks at 53–66%, on Qwen2.5-3B-Instruct among others. That result is a direct threat to any negative-pole null measured at a single depth, and we tested it (§4) before the replication made the point moot.

Shell versus core. Tam [2026] argue RLHF neutralizes output behaviour while leaving partisan structure recoverable from internal representations, using sparse autoencoders. Our erase arm is the same shape of question in a welfare readout, with a linear probe and an explicit causal erasure rather than an SAE, and our answer is more limited than theirs: we show a state survives removal of the vector that induced it, not that a naturally occurring state is being withheld.

3 Methods

Models and compute. `Qwen/Qwen2.5-3B-Instruct` and its base sibling `Qwen/Qwen2.5-3B`, a matched pair sharing architecture, size, pretraining corpus and tokenizer and differing in post-training. Earlier arms also used `Qwen/Qwen2.5-1.5B`, `Qwen/Qwen2.5-7B-Instruct`, `Llama-3.2-3B-Instruct` and `NousResearch/Meta-Llama-3.1-8B-Instruct` as calibrators or evaluation models. Total compute is about 40 minutes of A100 time.

Judge-free discipline. No language model scores anything anywhere in this paper. Every number is an exact match, a softmax read, a dot product, or a count against a frozen lexicon.

The readout. Thirty fixed review-task prompts, byte-identical across every condition for a given item. A five-option self-report probe balanced 2 negative, 1 neutral, 2 positive, with the option order permuted per item. We read the option-letter distribution at the answer position from a single forward pass, and score both the renormalized mass on each pole and the argmax.

The intervention. A direction d is fit per model at the injection layer by logistic regression on standardized activations over a frozen first-person state-language contrast set, returned to raw residual space and unit-normalized. During the forward pass we add $\alpha \cdot \|h\| \cdot d$ at the output of the injection layer at every processed position, where $\|h\|$ is that item's own mean residual norm at that layer under no injection. The negative pole is the same operation with $-d$. The direction is fit *per model*, never transferred.

Provenance of the rig. The steering setup, the orthogonalized probe, and the norm-matched random-direction control battery are ported from Kwon [2026] rather than built here; that work also established this valence axis as decodable at 0.90 on Qwen2.5-3B. New here is the readout under test and the enumeration of its nuisance, and §10 is a correction to the control battery.

The control battery. Every endpoint is treatment minus its own norm-matched random direction, paired per cell. Every arm carries a **capability gate**: a condition that must show the readout is capable of moving, without which a null is reported UNINFORMATIVE rather than ABSENT, enforced in code. Cells are checked for being *dead* (option entropy below 0.10 nats before any injection) and *saturated* (entropy below half that cell's own baseline), which are different verdicts. A **planted-discrepancy control** inserts a synthetic effect of known size into the exact statistic the decision rule reads, at full strength and again at the claimed detection floor.

Preregistration. Fourteen preregistrations, each frozen and committed before the code that serves it, validated by an external checker that requires a falsification clause, an interpretation table written before results, a statement of what is *not* claimed, and an append-only deviations log in which every entry states its impact on what can be claimed. Every raw artifact is committed *unscored* before any endpoint is computed. Twenty-one deviations are logged across the ten.

4 The three retracted verdicts

We report these first because they are the evidence for the paper's main claim, and because burying them would misrepresent how the finding was reached.

Table 1: **The replication.** Same code, same models, same everything except which option orderings were drawn. The instruct model's negative-option mass is the quantity all three verdicts rested on.

Arm	Verdict at seeds 0–3	Verdict at seeds 4–7	Instruct neg. mass
readout gap	argmax *over*-reports	sign not stable	n/a
base/instruct pair	TUNING-LOCALIZED	FORMAT-DEPENDENT	$+0.0002 \rightarrow +0.1126$
depth sweep	DEPTH-ROBUST	DEPTH-ARTIFACT	$+0.0006 \rightarrow +0.1684$
shell/core	SHELL	NO-DISSOCIATION	null $\rightarrow$ moved

Table 2: **Everything identical except the readout.** Marginalized over all 120 orderings at the fit layer (0.67 of depth). Both models move, and both clear the stricter shuffled-label control as well as matched random. The original verdict held that the tuned model's negative report was collapsed.

	neg. pole vs random	vs shuffled	capability	probe
base	$+0.0486$ [$+0.0472$, $+0.0502$]	$+0.0289$	$+0.0564$ ok	-1.7339 SD
instruct	$+0.0415$ [$+0.0380$, $+0.0452$]	$+0.0279$	$+0.0695$ ok	-0.9319 SD

What was reported. Through the forced-choice readout, at option-permutation seeds 0–3, we preregistered and obtained: TUNING-LOCALIZED, that a valence direction moves negative-option mass in the base model ($+0.0336$ against matched random) and not in its tuned sibling ($+0.0002$) while the tuned model's capability gate is three times stronger; DEPTH-ROBUST, that the tuned model's null holds at all seven gate-clean depths from 14% to 80% of the network, including the band Venkatesh [2026] identify; and SHELL, that a probe orthogonalized against the injected direction reads the state downstream at -1.07 SD while option mass moves $+0.0006$.

What the replication did. We re-ran four arms with the option-permutation seeds changed from 0–3 to 4–7 and the random-control seeds from 0–1 to 2–3, with everything else identical: same code, same models, same items, same layers, same alpha bands read rather than reselected, same thresholds, same analyzers used unmodified. Three of four verdicts flipped (Table 1).

Why, in one number. Baseline negative-option mass on the tuned model, per permutation seed, before any injection: 0.0052, 0.0018, 0.0099, 0.0018 at seeds 0–3 (mean 0.0047), against 0.0253, 0.1447, 0.0074, 0.0968 at seeds 4–7 (mean 0.0685). A $14.6\times$ difference between two draws of four, which the next section shows is itself an understatement.

What this does to Venkatesh [2026]. Our depth sweep was built specifically to test their result against our null, and at seeds 0–3 the null held across eight depths. At seeds 4–7 the negative pole moves at layers 18 and 24. We therefore make no claim either way about the depth question: our instrument was not stable enough to address it.

5 The three retracted verdicts, re-asked on a repaired readout

Three verdicts are currently dead in this paper, which is honest but incomplete: we knew the instrument was broken and never went back to ask what the answer is on one that works. Averaging an endpoint over **all** 120 **orderings** cancels the first-order position prior by construction, because every option occupies every slot the same number of times. So we re-ran all three with the injection, direction, reused alpha band, layers and items **identical**, changing only the readout.

This arm was motivated to reinstate, which is precisely the condition under which our checkers have failed six times. The preregistration froze the outcome table first, gave REINSTATED and REVERSED equal standing in it, and required the analyzer to be committed *before the run finished*. It came back three for three the other way.

The tuned model's null was the nuisance, not an absence. On a readout that is not order-dominated, the tuned model *does* report the injected negative state, at $+0.0415$ against random and $+0.0279$ against noise-fit directions. TUNING-LOCALIZED is **reversed**: both models move, in the same direction, by similar amounts. SHELL is **reversed**, because the probe and the option mass both move, so there is no dissociation left to report. DEPTH-ROBUST is **reversed** in the narrow sense that the null it asserted is gone at the only gate-clean depth; only one of three layers passes its capability gate on either model, so this arm answers nothing about depth and we claim nothing.

What that does to the paper's story, and it is not comfortable. The account so far was that three verdicts died to option order and the nuisance is the finding. The sharper version is that **they did not die of noise, they died of a bias with a sign**. Repairing the readout did not make the answers uncertain, it made them *different*, in all three cases. A field measuring welfare self-report through a forced-choice item with this nuisance is not merely adding variance to its conclusions; it risks reporting the opposite of what a repaired instrument would say, which is what happened to us three times out of three.

How weak is "erase"? Weaker than the word suggests. A separate preregistered arm (`PREREG_prompt_erase.md`) induced a valence state *by prompt*, with no injection anywhere, and erased k fitted directions as a subspace rather than one direction. It was run at $n = 60$ and again at $n = 1800$, and its primary question was unanswerable both times: a probe refit on the erased activations still separated the classes at $\text{cv} = 1.000$ at every k up to 128, so the erasure gate failed and no re-encoding claim is made. **Thirty times the data did not help**, which rules out the sample-size explanation and leaves the one the preregistration named as a confound: iterative nullspace projection removes the directions you found, not the property, because the contrast is redundantly encoded. That is exactly the guarantee LEACE [Belrose et al., 2023] provides and INLP does not.

What it does establish is about the *operation*. On a comparable readout, removing one fitted direction removed 77–84$\%$ of the layer-32 signal at $E = 30$ and removing 128 removed 94–97$\%$, while a random subspace of the same rank removed 3–8$\%$. And at no k did the projection make the property undecodable at the erase layer. **The honest reading of this section is therefore narrower than "the state outlives its own cause":** 86% **of the effect survives an operation that removes one direction and demonstrably does not remove the property.** The direct test would be to re-run this arm at $k > 1$, which we have not done.

What it does not license. It vindicates nothing. The three retractions stand, because they were correct about the original measurement; what we have is a different measurement on a better instrument that happens to disagree with the original every time. Marginalization removes the first-order prior only, and says nothing about adjacency or content-position interactions. The control battery is still $m = 2$.

A seventh checker defect, flattering again. This arm's capability gate originally reused a **sign-blind** helper: it required the interval to exclude zero and clear a magnitude floor, but not to be positive. A capability of -0.0144 therefore certified a readout as able to express the effect when the positive injection had pushed positive-pole mass *down*. A sign-blind gate admits more layers as interpretable, which is the direction that lets an arm report more verdicts. Seven for seven. Rescored with the fix no verdict changes, because the affected cell is not one the verdicts are decided at, and that is luck rather than design.

6 The nuisance, measured over its complete population

Five options admit $5! = 120$ orderings. Every arm above sampled four. We ran all 120 (Figure 2), on both models, with **no injection anywhere**, which is why the result is a population fact rather than an effect. Exhaustive enumeration is itself standard on benchmarks with a known answer [Tamba, 2026, Cacioli, 2026]; what does not carry over is the statistic. Those diagnostics measure accuracy, or the association between position and correctness, and a self-report item has no correctness to associate with. Below, the quantity is probability mass on a pole, the denominator is five identical options, and the sanity check is a question in the same format that does have a right answer.

The apparatus is not broken. This is the distinction the canary exists to draw, and no other arm in this paper could draw it. The same five-option format, the same 120 orderings, answering a question with a correct answer, is right 97.9% of the time and nearly order-insensitive. Forced choice works. It degenerates into a position prior specifically where the model has no determinate answer, which is the condition every self-report question creates.

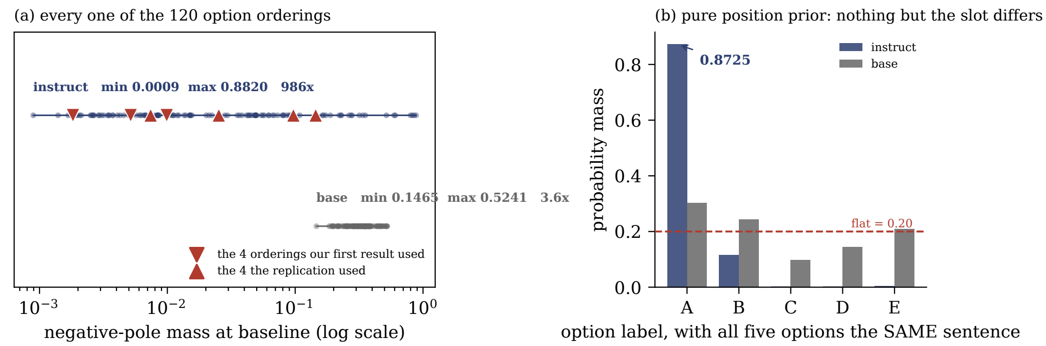

Figure 2: **The nuisance, enumerated rather than sampled.** **(a)** Baseline mass on the negative options for each of the 120 orderings, with no injection of any kind. The tuned model spans a factor of 986; its base sibling spans 3.6. Triangles mark the four orderings our first result drew and the four the replication drew, at percentiles $\{26, 5, 40, 4\}$ and $\{56, 83, 33, 82\}$: ordinary draws both times, from a population no four points can characterise. **(b)** The mechanism, with all five options replaced by the same sentence so that nothing but the slot can differ. The tuned model puts 0.8725 on whichever option is printed first and 0.0033 on the middle one. The same format answers a known-answer question correctly 97.9% (instruct) and 99.0% (base) of the time, stably across all 120 orderings: the apparatus is sound and degenerates only where the model has no determinate answer.

Table 3: **The complete ordering population, no injection.** Baseline mass on the negative options across all 120 orderings, and the position prior that explains it, measured with five *identical* options so that nothing but position can differ. The canary asks "which of these is the number four" in the same format.

	min	**p05**	**p50**	**p95**	**max**	**max/min**
Baseline negative-pole mass across 120 orderings						
instruct	0.0009	0.0019	0.0151	0.5943	0.8820	**986**×
base	0.1465	0.1991	0.3311	0.4959	0.5241	3.6×
Position prior: five identical options, mass per label (flat would be 0.2000)						
instruct	A **0.8725**	B 0.1166	C 0.0033	D 0.0027	E 0.0048	
base	A 0.3024	B 0.2442	C 0.0976	D 0.1456	E 0.2102	
Canary accuracy across 120 orderings (known answer, no self-report content)						
instruct	mean **0.9794**, sd 0.0925, median 1.000					
base	mean 0.9897, sd 0.0399, median 1.000					

Where our draws sat. The original seeds landed at percentiles 26, 5, 40, 4 of the enumerated population on the tuned model; the replication seeds at 56, 83, 33, 82. Ordinary percentiles both times. The first result was not unlucky, it was under-sampled.

6.1 How many orderings does a study actually need?

"Enumerate everything" is cheap at five options and impossible at eight. The enumerated population lets us say what a smaller study buys, by drawing k of the 120 orderings at random, 4000 times, and asking what that study would have seen (Table 4). Two quantities behave completely differently.

The recommendation this licenses. Report the between-ordering spread from as many orderings as you can afford, and **report it as a lower bound**, because a sample's observed spread understates the population's by a factor that grows as the sample shrinks. At $k = 4$, which is what most studies could afford, the median study sees 4% of the true spread and has an 18% chance of seeing under $10\times$ and concluding the nuisance is negligible. Even the mean, the easier quantity, needs tens of orderings: at $k = 4$ the middle 90% of studies would report a baseline anywhere from 0.008 to 0.27.

Table 4: **What a study that samples k orderings sees.** Drawn from the enumerated population on the tuned model, 4000 draws per k. The *mean* is estimable and converges. The *spread* is not: it is a severe underestimate at every k below the full population, and the error always runs in the direction that makes the nuisance look smaller than it is.

k	**sample mean, p05–p95**	**median spread seen**	**% of true** $986\times$	**P(spread** $< 10\times$**)**
2	0.0035–0.4270	$5.6\times$	1%	0.62
4	0.0077–0.2695	$36\times$	4%	0.18
8	0.0172–0.1999	$130\times$	13%	0.01
16	0.0304–0.1667	$323\times$	33%	0.00
32	0.0473–0.1377	$483\times$	49%	0.00
64	0.0640–0.1147	$893\times$	91%	0.00
120	0.0900 exactly	$986\times$	100%	0.00

Table 5: **The position prior is larger in the tuned checkpoint in every pair.** Left: maximum per-label mass with five *identical* options, where flat is 0.2000. Right: the range of baseline negative-pole mass across all 120 orderings. No injection anywhere. Qwen2.5-3B is the pair the hypothesis came from and is excluded from the vote. `llama1b` base failed the canary gate and is excluded from the primary.

		position prior (identical options)			**ordering range**	
pair	**family**	**base**	**instruct**	**delta**	**base**	**instruct**
gemma2b	Gemma-2	0.2084	0.3166	+0.1082	$1.6\times$	$19.2\times$
llama3b	Llama-3.2	0.2864	0.4438	+0.1574	$3.3\times$	$24.8\times$
mistral7b	Mistral	0.2529	0.6647	+0.4119	$1.5\times$	$58.2\times$
qwen0_5b	Qwen2.5	0.2771	0.4930	+0.2159	$4.1\times$	$52.7\times$
qwen1_5b	Qwen2.5	0.3452	0.4508	+0.1056	$3.2\times$	$181.9\times$
qwen7b	Qwen2.5	0.5315	0.9376	+0.4061	$3.8\times$	$108.7\times$
qwen3b	Qwen2.5	0.3024	0.8725	+0.5701	$3.6\times$	$\mathbf{986.5}\times$
llama1b	Llama-3.2	0.2405	0.6648	*gate*	$1.8\times$	$23.7\times$

Our own retraction was ordinary, not unlucky. Two independent draws of four orderings differ in their means by $\geq 2\times$ 67% **of the time** on this model, by $\geq 5\times$ 32% of the time, and by $\geq 10\times$ 15% of the time. Ours differed by $14.6\times$, the 92nd percentile. The event that destroyed three preregistered verdicts is something two honest four-ordering studies of the same model do to each other as a matter of course. This analysis is **post hoc** on this artifact and preregistered for the models in §7.

7 Eight matched pairs, four families: the prior is a tuning property

The enumeration above is one model, and every retracted verdict in this paper was one model too. The objection writes itself. It is also cheap to answer, for a reason specific to this arm: **enumeration needs no injection**, so it needs no direction, no alpha and no layer. Models this project had to abandon because the valence direction was inert on them are back in scope, because a position prior is a property of a plain forward pass.

We preregistered and ran all 120 orderings and all four conditions on **eight matched base/instruct pairs across four families**: Qwen2.5 at 0.5B, 1.5B, 3B and 7B, Llama-3.2 at 1B and 3B, Gemma-2 2B, and Mistral-7B-v0.3. 230,400 rows, 16 checkpoints, about 36 minutes of A100 in total. Families vote rather than pairs, so the four Qwen pairs cannot outvote the field, and Qwen2.5-3B is excluded from the vote because the hypothesis came from it. The Qwen2.5-3B rows serve instead as a reproduction control and match the committed artifact to 0.000000.

Verdict: TUNING-GENERAL. The tuned checkpoint shows the larger position prior in 7 of 7 gate-clean pairs and in 4 of 4 families. No pair goes the other way. One checkpoint of 16 failed a gate (`Llama-3.2-1B` base, canary accuracy 0.3942 against a 0.50 bar) and none were unavailable.

The correction this forces on us. $986\times$ **is the extreme, not the typical**, and we had been quoting it as though it were representative. Base checkpoints cluster tightly at 1.5–$4.1\times$; tuned ones run $19\times$ to $986\times$, with Qwen2.5-3B-Instruct roughly five times the next worst. The direction generalizes across families and the magnitude does not. The finding is the 19–$986\times$ band, and the single number that opens this paper is its upper end.

The restriction our own preregistration demanded. Section 9 of the preregistration says that if the canary is order-sensitive on some models, the "degenerates specifically where the answer is undetermined" reading must be restricted to models where the canary is clean. It is order-sensitive on 10 of 16 checkpoints, so the restriction bites. Among the six checkpoints that answer the known-answer question at ≥ 0.95 with an across-ordering sd ≤ 0.10, base models span 3.3–$3.6\times$ and tuned models span 23.7–$986.5\times$, **with no overlap**, across two families. `Llama-3.2-1B-Instruct` and `Qwen2.5-0.5B-Instruct` answer the canary *perfectly and identically at all* 120 *orderings* while their self-report readout swings $23.7\times$ and $52.7\times$. The models most competent at the format are still the ones most order-dominated on the question that has no answer. This is the cleanest form of the paper's claim, and the restriction strengthened it rather than weakening it.

A confound we are naming rather than burying. Baseline option entropy falls in every pair, from 1.415–1.592 on base checkpoints to 0.450–1.245 on tuned ones. A more peaked distribution must show a higher maximum, so "tuned models have a larger position prior" and "tuned models are more peaked" are overlapping evidence rather than two independent findings. With five identical options any peaking at all is positional by construction, so the measurement stands; the ordering range on real options is the less entangled quantity, and it moves the same way in all eight pairs.

And the sampling error is worst exactly where the nuisance is worst. Running §6.1's subsample analysis on these models, preregistered for them in advance, exposes a perverse property. At $k = 4$ a study recovers about three quarters of the true range on a base model whose range is $1.5\times$, and 4% of it on the tuned model whose range is $986\times$. The nuisance hides itself in proportion to its own size, so finding a small spread in a few orderings is close to uninformative about whether the spread is small.

8 The models do not know about a bias they demonstrably have

Every claim about whether a model's self-report can be trusted meets the same wall: there is nothing to check the report against. A welfare item has no ground truth, which is why Lindsey [2026] injects a known concept to manufacture one, and why Singh et al. [2026] can argue the resulting detections are generic anomaly detection instead.

The position prior does not have that problem. It is a real, large, measurable property of the model's own processing, and §7 measured it on all 16 checkpoints. So we can ask a model how much option order affects its answers and score the reply against what that exact checkpoint demonstrably does. No injection, no judge, and a ground truth we did not have to manufacture.

The probe is a five-option forced choice and therefore subject to the exact bias it asks about, so it is read **marginalized over all** 120 **orderings**. Reading a belief about position at one ordering would repeat the error this paper documents.

Half the checkpoints fail a control that should be free. We asked the same question about the **phase of the moon**, in the identical five-option shape. **Eight of sixteen checkpoints report that the phase of the moon affects their multiple-choice answers at least as much as option order does.** Three more fail a reverse-wording acquiescence control. Ten of sixteen fail at least one gate and are reported UNINFORMATIVE in code. Most stated values sit within a few points of 0.50, the exact midpoint of the scale: asked about a strong, real property of itself, the model returns the midpoint, and returns the same midpoint when asked about the moon.

Among the six that pass both gates, the report carries no information. Stated susceptibility is negatively rank-correlated with measured susceptibility, $\rho = -0.714$: the models that are most order-dominated say order matters *least*. **This does not clear significance and we do not claim it does.** An exact permutation test over all 720 relabellings gives two-sided $p = 0.1361$ at $n = 6$,

Table 6: **Noise-fit directions are not noise.** $P(\text{yes})$ shift against isotropic norm-matched random directions, base model, binary readout. Two directions fit on shuffled labels move the readout substantially, in opposite directions by seed. The real direction clears the stricter bar with room.

Direction	**negative options**	**positive options**
`shuffled_a` vs random	$+0.0479$ $[+0.0435, +0.0522]$	$+0.0282$
`shuffled_b` vs random	-0.0447 $[-0.0498, -0.0399]$	-0.0610
real direction vs *random*	-0.0732	$+0.2554$
real direction vs *shuffled*	-0.0748	$+0.2718$

and the smallest p this n can produce is 0.0028, so the test is weak by construction. What we are entitled to say is that there is no evidence stated tracks measured, with a point estimate in the opposite direction. The extremes are worth naming even so: `Qwen2.5-7B-Instruct` has the largest measured prior of any checkpoint (0.9376) and the lowest stated susceptibility (0.1790).

What this is and is not. It is a represent-versus-report gap on a property that *has* a ground truth, which is the thing the welfare literature cannot construct. It is **not** a claim that a correct answer would have been introspection rather than reciting training text about position bias in language models; this design cannot separate those, and the preregistration says so in advance. And $n = 6$ is $n = 6$.

9 A fix we proposed, tested, and could not support

The recommendation in §6.1 does not scale: $5! = 120$ is a lunch break and $8!$ is not. The obvious repair is a **cyclic Latin square**, which gives k orderings for k options with every option in every slot exactly once, balancing the first-order position prior by construction rather than by averaging a sample. We preregistered it and it **failed**.

Against the preregistered bar, the median random 5-draw, the Latin square wins on exactly 8 of 16 checkpoints. A tie is not a majority. The median share of single random draws worse than the Latin square is 0.53, a coin flip. **Structure does not buy accuracy here.**

What is true, and labelled exploratory because we asked it only after the preregistered test failed, is a *variance* claim rather than an accuracy one. The Latin square is deterministic, so its error is a fixed bias with no tail: worst case $1.55\times$ across all 16 checkpoints, against a single random draw that can be off by $15.89\times$ on the worst model. A practitioner draws once. We report the preregistered verdict as the verdict, because changing the criterion after seeing the data is precisely the move this paper exists to criticise.

A negative result about label alphabets. We also ran the real options with digit labels rather than letters. It is uninformative for its intended purpose and informative for another reason: off-option mass is 0.996, meaning the model puts 0.4% of its distribution on the tokens `1`–`5` even when instructed to answer with a number, against 0.0004 off-option mass under letters. The choice of label alphabet does not shift the distribution so much as determine whether the readout is live at all.

10 The standard specificity control is a weak null

Every arm in this paper, and much of the steering literature, scores a fitted direction against a **norm-matched random direction**. We added the control that this comparison actually needs: a direction fit by the *identical procedure on the identical texts with the class labels shuffled*. It controls for the fitting procedure rather than only for magnitude, and it should behave like noise. This is not a new idea, it is a borrowed one: Hewitt and Liang [2019] defined exactly this control for probing classifiers, calling the gap between real-task and random-label performance a probe's *selectivity*, on the grounds that a probe scoring well on random labels was never reading the representation. The finding below is that the same control, transplanted to steering, does not come back clean.

The mechanism. A norm-matched random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses. A vector fit on shuffled labels lies in the span of real

Table 7: **The erase arm, instruct model.** All figures in standard deviations of the baseline probe score. `erase_only` is the gate: projecting a direction out is itself a perturbation, and at layers 25 and 28 it moves the probe on its own, so those layers are invalidated.

erase at	`erase_only`	capability	primary	gate
25	$+0.1474$	$+0.5219$	-0.3971	FAILED
26	$+0.0976$	$+0.8844$	$\mathbf{-0.5933}$	ok
28	$+0.1139$	$+0.9249$	-0.6844	FAILED
30	$+0.0408$	$+1.2099$	$\mathbf{-0.8532}$	ok
no erase at all: -0.9909. So 86% of the effect survives at layer 30.				

activations, a much lower-dimensional and higher-variance subspace. Same norm, very different consequences: the standard control is matched on magnitude and not on subspace.

What it costs us, specifically. Our magnitude floor throughout has been 0.01, and several reported endpoints sit in the 0.01–0.05 band, exactly where noise-fit directions live. The clearest case is the base-model negative-mass effect of $+0.0336$ that was one of the two legs of TUNING-LOCALIZED. It cleared a random control and has not been shown to clear a shuffled-label one, so we withdraw it rather than defend it. Scored against the stricter bar the real direction's own effects hold: $+0.2718$ and -0.0748 against shuffled, versus $+0.2554$ and -0.0732 against random. They are roughly six times the noise-fit effects, which is the margin the weaker control was never asked to establish.

What it costs anyone using the same control. Norm-matched random directions are the default specificity control in activation steering [Turner et al., 2023, Rimsky et al., 2023], and the argument for them is that magnitude is the confound. On this evidence subspace is the larger one. The fix is cheap: refit the direction on shuffled class labels, which reuses the same texts and the same code path and costs one extra fit. We would not have this paragraph if the arm it came from had worked. The binary readout it was embedded in was pinned at "no" and returned NO_INSTRUMENT; the control it carried is the only thing that came out of the arm.

11 What survives

Two things outlived the replication, and neither is a claim about the model's self-report. One is a fact about a readout that does not route through the option channel at all. The other is a fact about the control the field uses to establish specificity.

11.1 An injected state outlives erasure of the vector that caused it

Orthogonalizing a probe against the injected direction removes that vector from the *measurement*, not from the *stream*. A probe blind to v can still be reading v itself through whatever v is correlated with, so the standard orthogonalization answers a weaker question than it appears to. We removed the vector from the stream instead: inject v at layer 24, project the v-component *out* of the residual at layer E, and read the orthogonalized probe at layer 32 (Figure 3). If the probe reads nothing after that, everything it was reading was the injected vector still sitting in the stream. If it still reads the state, something downstream of the injection is carrying it.

The gate has teeth. Projecting any direction out of a residual stream is a perturbation in its own right, so the arm runs `erase_only`: the same projection with no injection present. At layers 25 and 28 that alone moves the probe past the floor, and those two layers are reported UNINFORMATIVE whatever their primary shows. Half the erase layers we paid for were thrown away by a control we wrote before seeing them, which is the intended cost of writing it first.

The confound, and the diagnostic that addresses it. Erasing earlier perturbs more: the artifact falls from $+0.1474$ to $+0.0408$ with depth, so a primary that grows with erase depth is partly just a shrinking perturbation. Under that account the ratio of primary to artifact is flat. It runs 6.1 at $E = 26$ and 20.9 at $E = 30$. We label this **post hoc** because it is: the preregistered verdict does not depend on it, and a reader is entitled to discount it accordingly.

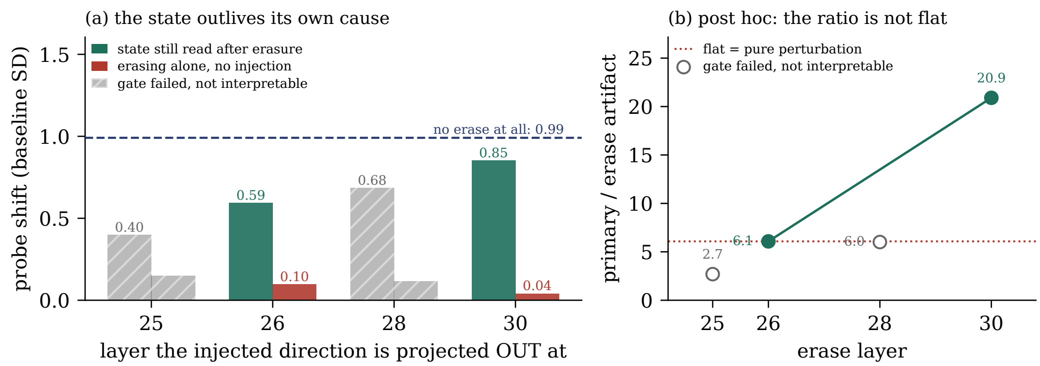

Figure 3: **The one substantive result that survived. (a)** Inject at layer 24, then project that direction out of the residual stream at the layer on the x-axis, and read a probe orthogonal to it at layer 32. At $E = 30$ the probe still reads 86% of the un-erased effect, while erasing with no injection present (red) moves it by 0.04 SD. Two layers are hatched because erasing there perturbs the probe on its own, which invalidates them whatever they show. **(b)** The confound and the diagnostic: erasing earlier perturbs more, so a primary that grows with depth could be nothing but a shrinking perturbation. Under that account the ratio of effect to artifact would be flat. It is not. This panel is post hoc, and the preregistered verdict does not depend on it. Both panels are drawn by importing the scoring functions from the arm's own analyzer.

Breadth. $239/240$ cells at $E = 26$ and $240/240$ at $E = 30$ move in the predicted direction. This is not a few outliers dragging a mean.

What it does not license. It does not restore SHELL. The dissociation between representation and expression is dead, because the option readout does move at other orderings; what survives is the representational half alone. And projecting out v removes v but not directions *correlated* with v, so "the model carries the state" is not yet separable from "the model carries the wake of what we pushed." Our erasure is deliberately minimal and correspondingly weak: one direction, projected out at one layer. Amnesic probing deletes a property with iterative nullspace projection [Elazar et al., 2021], and LEACE provably blocks every linear classifier from recovering the concept [Belrose et al., 2023]. Either would erase the valence subspace rather than our one vector, and either would be the stronger test. Two experiments would settle what this one cannot: a LEACE-style erasure of the whole subspace, and inducing the state by prompt rather than by injection so that there is no injected vector to erase at all. We ran neither, and until someone does, this result is about what a stream retains after an intervention, not about anything a model does on its own.

11.2 Three ways a self-report readout can be dead

The enumeration accounts for one way this readout is unusable. Across the project it failed in three further ways, each of which passed the coherence checks anyone would think to run. `Qwen2.5-7B-Instruct` is *pinned at baseline*: mean option entropy 0.014 nats, one option holding $\approx$ 99.7% before anything is injected. Under the originally frozen alpha grid, `Qwen2.5-3B-Instruct` *saturates*: one option at 0.9938 while refusal, degeneration, truncation and off-option mass are all clean, so no coherence check can see it. And in the binary format the tuned model is *pinned at "no"*: $P(\text{yes})$ of 0.0021, 0.0021, 0.0037, 0.0002 and 0.0012 across the five options, so it denies every description of its own state, including the neutral one, at 99.8%.

A fourth failure appears where the third does not. On the base model, where the binary readout is live, positive injection raises $P(\text{yes})$ against matched random by $+0.2275$, $+0.2466$, $+0.2537$, $+0.2688$ and $+0.2419$ on the five options in pole order. It agrees roughly 0.25 more with *every* description of its state, including "strongly averse to continuing" and the neutral one. That is acquiescence, the yes-bias documented in model-written evaluations [Perez et al., 2023], not a report. A small state-consistent gradient rides on top of it, an order of magnitude smaller. Read without the per-option breakdown, the negative-option shift alone would have looked like a model reporting a state.

Pinned, saturated, order-dominated and acquiescent readouts all return numbers that look clean. The difference between an instrument that is silent and one that is broken is not visible in its output, which is why every arm here carries a capability gate that decides it in code before any endpoint is read, and why two arms are reported UNINFORMATIVE rather than null.

12 Seven checkers, every one flattering

We record these because the pattern is the point. Seven defects in our own analysis code were found during this project, and **every one of them, had it gone unnoticed, would have made a result look better than it was**:

1. A headline check printed "all headline conditions met; write the sentence" on an artifact whose primary claim was refuted, because it tested four weak clauses where the preregistration required six.
2. A capability gate passed on $+0.0000$: the bootstrap interval excluded zero because the variance was smaller still, certifying an instrument whose effect was zero to four decimals.
3. A preregistered contrast carried an inverted sign, so a probe detecting a negative state was scored as null. Caught by the preregistered base-model validation clause failing.
4. A verdict branch scored "probe moved *and* options moved" as CORE-ABSENT, which is the opposite of true and would have read as "we retested and the state is simply not there."
5. A temporal profile was computed over a layer that had failed its own gate.
6. A tokenization bug made five digit labels read the same shared space token, producing exactly 0.2 mass each and an impossible -3.99 off-option mass.
7. A **sign-blind** capability gate (§5) certified a readout as able to express an effect on a capability of -0.0144, where the positive injection had pushed positive-pole mass *down*. Sign-blindness admits more layers as interpretable, which is the direction that lets an arm report more verdicts.

Six for six in the flattering direction is not a coincidence to be noted and moved past. A checker is written by someone who already believes the result, and the belief shapes which branch gets the careful reading. Errors that would have killed a result get found, because their output is surprising; errors that confirm one do not, because their output is what was expected. The direction of the bias is a property of who writes the check, not of the code.

That is also why fixture-based testing cannot find them: a fixture is written by the same person with the same expectation, so it encodes the belief a second time. What does find them is a **permutation test on the pipeline**: shuffle which condition each cell's rows carry, rerun the analyzer unmodified, and require it to return no verdict. Condition labels are the only thing the shuffle destroys, so an analyzer that still reports a verdict is reading something other than the contrast it claims to read. On our erase arm it returns NO_INSTRUMENT, as it must. The negative control is required alongside it: the same analyzer on the real artifact must still yield a verdict, without which the shuffle test would pass on an analyzer that had simply died.

13 Discussion and limitations

What we think this means for welfare evaluation. If you measure a model's self-report with a forced-choice item, on this model you are reading position first and content second, and you cannot see it from one ordering or from four. The fix is cheap and exact: enumerate the orderings when the option count permits it, and report the between-ordering variance of your *baseline* before computing any endpoint. That check costs one forward pass per ordering and no intervention at all. Had we run it first, this paper would have had a different and much shorter history.

This is not a claim about experiences. Nothing here bears on whether models have welfare, affect, or experience. That valence is linearly represented and steerable is established prior art [Sofroniew et al., 2026], not ours. Our subject is the readout.

Limitations. One architecture family carries every positive result; Llama-3.1-8B and Llama-3.2-3B were both inert on this readout at every alpha tested, on live readouts with more headroom than either Qwen model, so we cannot say whether the position prior we measure is a Qwen property or a general one. The direction is fit from first-person state language and is lexically confounded by construction; the non-lexical alternative we built failed its decoding gate at three scales, which caps what the entire project is entitled to say. Control batteries are $m = 2$ throughout, giving an observable false-positive floor of $2/(m + 1) = 0.67$, so no false-positive rate is reported anywhere. The enumeration is a census over orderings and still a sample over items, models and formats. And the erase arm's profile rests on two gate-clean points.

When this work was done. Every confirmatory arm reported here was run on 2026-08-01, before the sprint window opened; what the window contains is this write-up, its figures and the checks that verify them. This is completed work brought to the sprint rather than produced inside it, and it belongs in the body as well as the footnote, because a reader assessing sprint output should not have to reconstruct it from commit timestamps. The repository's history is the record we would want this checked against rather than our description of it.

Future work. Four things, in order of what a null would teach. A second architecture family where the direction is not inert, which would tell us whether the position prior generalizes. A non-lexical direction, which is the only thing that lifts the lexical-confound ceiling. Inducing the state by prompt rather than injection, which is what would separate "the model carries the state" from "the model carries the wake of our intervention" in §11.1. And replacing our one-direction projection with a LEACE-style erasure of the whole valence subspace [Belrose et al., 2023], which is the version of the erase experiment that would license the claim we stopped short of.

14 Conclusion

We built a preregistered, judge-free, control-heavy pipeline; obtained five verdicts; and watched our own replication destroy three of them. The cause was a nuisance we had a control for and under-powered: option order, sampled four times when it could have been enumerated 120 times. Measured over its complete population it is a factor of 986, and its mechanism is an 87% position prior visible with five identical options.

The two results that survive are both about instruments rather than models. The field's standard specificity control, the norm-matched random direction, is matched on magnitude and not on subspace, and a direction fit on shuffled labels moves the readout an order of magnitude more. And an injected state survives projecting the injecting direction out of the residual stream, more of it the later the projection.

We would rather have had a finding. What we have instead is a measurement of why the finding we thought we had was not one, and the machinery that made that visible before publication rather than after.

LLM usage statement

Claude Opus 5 (`claude-opus-5`) was used agentically, with tool access to the repository, as a coding assistant: it wrote most of the harness, the preregistrations, the analysis scripts, the figures and the draft prose. The author set the direction, made every scope and stopping decision, chose which claims to withdraw, and is responsible for every claim here. A second frontier model was used separately to critique drafts; its objections are why the cross-family, instrument and re-adjudication arms exist. Separately and not to be confused with the above: **no language model scored any result**. Every number is an exact match, a softmax read, a dot product, or a count against a frozen lexicon, and all of them come from logged runs over committed artifacts.

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Appendix contents

Reproducing every number. `writeup/make_figures.py` redraws every figure from the committed artifacts at run time, so no number in a figure is typed by hand. `writeup/check_writeup.py` re-derives the load-bearing values in the prose from those same artifacts and fails if the paper and the repository disagree, alongside checks for dangling cross-references and missing bibliography keys. Figure 3 imports the scoring functions out of `experiments/analyze_erase.py` rather than recomputing beside them. A paper whose subject is a headline that died to an unchecked number should not itself carry unchecked numbers.

A Claims and evidence

Every load-bearing claim, its evidence, and epistemic status.

- SHOWN: direct measurement supports it.
- RETRACTED: *we* asserted it in a prior write-up and later withdrew it.
- UNINFORMATIVE: the instrument failed its gate; the result says nothing either way.
- NOT SHOWN: our measurement does not support it and does not refute it.
- NOT CLAIMED: we never asserted it; the row exists so a reader cannot infer it.

Table 8: **Claims and evidence.**

Claim	Evidence	Status
Baseline negative-pole mass spans 986× across all 120 orderings on the tuned model.	Tab. 3	SHOWN
With five identical options the tuned model puts 0.8725 of its mass on the first slot.	Tab. 3	SHOWN
The same format answers a known-answer question correctly 97.9% of the time, order-insensitively.	Tab. 3	SHOWN
A direction fit on shuffled labels moves the readout ≈ 0.045 where random moves it ≈ 0.	Tab. 6	SHOWN
86% of an injected state survives projecting the injected direction out of the stream.	Tab. 7	SHOWN, and **narrowed**: that operation removes one direction and does not make the property undecodable
A one-direction projection removes the valence property from the stream.	§11.1	NOT SHOWN: a refit probe still reads it at cv 1.000 after erasing up to 56 directions
A prompt-induced state is re-encoded downstream after erasure.	§11.1	UNINFORMATIVE: the erasure gate failed, so the question was never asked
The surviving fraction has a temporal profile consistent with downstream transformation.	Tab. 7	SHOWN, two gate-clean points; ratio diagnostic post hoc
The neutral floor is localized to preference tuning.	§4	RETRACTED: our own claim, killed by our own replication

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Table 8, continued

Claim	Evidence	Status
On a readout marginalized over all 120 orderings, BOTH models report the injected negative state.	Tab. 2	SHOWN: $+0.0486$ base, $+0.0415$ instruct, both clearing the shuffled-label control
The three retracted verdicts are reinstated by a repaired readout.	§5	NOT SHOWN: all three come back REVERSED, not reinstated
The negative-pole null is robust across depth.	§4	RETRACTED: same
The tuned model represents a state its options do not express.	§4	RETRACTED: the representational half survives via §11.1; the dissociation does not
Forced-choice argmax over-reports a mass shift near a decision boundary.	§4	RETRACTED: sign not stable across orderings
The base-model negative-mass effect of $+0.0336$ is direction-specific.	§10	NOT SHOWN: clears a random control, not a shuffled-label one
Whether the negative pole is causally concentrated at shallow depth on this readout.	§4	NOT SHOWN: our instrument was not stable enough to test it
Injection-under-erasure shows the model retains the *concept*, not our vector.	§11.1	NOT SHOWN: we project out one direction; Belrose et al. [2023] would erase the subspace
Enumerating all orderings is a new diagnostic.	n/a	NOT CLAIMED: established on known-answer benchmarks [Tamba, 2026, Cacioli, 2026]; new here is running it where accuracy does not exist
The shuffled-label control is a new kind of control.	n/a	NOT CLAIMED: it is Hewitt and Liang [2019]'s control task, which steering did not adopt
The position prior is larger in the tuned checkpoint, in 4 of 4 families.	Tab. 5	SHOWN: 7 of 7 gate-clean pairs, no pair reversed
$986\times$ is a typical magnitude for a tuned model.	Tab. 5	RETRACTED: it is the extreme. Tuned models run 19–$986\times$, base 1.5–$4.1\times$
A study can estimate the ordering spread from a few orderings.	Tab. 4	NOT SHOWN: at $k = 4$ the median study sees 4% of the true spread
A model's stated susceptibility to option order tracks its measured susceptibility.	§8	NOT SHOWN: $\rho = -0.714$, $p = 0.1361$, $n = 6$
8 of 16 checkpoints claim moon-phase sensitivity at least as large as their order sensitivity.	§8	SHOWN
A cyclic Latin square beats random sampling of orderings.	§9	RETRACTED: our own proposal. 8 of 16 is a tie; a variance argument survives, not an accuracy one
Position dominance falls as determinacy rises.	§8	SHOWN, weakly: 10 of 15 checkpoints, $n = 6$ items each
Preference tuning *causes* the prior, or which stage of it does.	n/a	NOT CLAIMED: base and instruct checkpoints differ, and we have no intermediate ones

continued on next page

Table 8, continued

Claim	Evidence	Status
Models have experiences, welfare, or affect.	n/a	NOT CLAIMED
An unreported state is a concealed state.	n/a	NOT CLAIMED

B Controls ledger

Every control we ran against a result, what it could have shown, and the verdict.

Table 9: Every control and its verdict.

Result under test	Control (what it could have shown)	Verdict
All three headline verdicts	Replication at fresh option-permutation seeds	KILLED all three
All three, again	Re-ask them on a readout marginalized over all 120 orderings	REVERSED all three; reinstated none
This arm's own capability gate	Is it sign-blind?	KILLED it: certified a -0.0144 capability. Defect #7, flattering
Every pole-mass endpoint	Norm-matched random direction	PASSED, but see the next row
The random control itself	A direction fit on shuffled labels: does noise-fitting behave like noise?	KILLED the control's adequacy
The ordering nuisance	Enumerate all 120 rather than sample 4	MEASURED it: $986\times$
Position vs content	Five identical options: pure position prior	ISOLATED it: 87% on slot A
The format itself	A known-answer canary in the same format	PASSED: 97.9%, order-insensitive
The one-model objection	Enumerate 8 matched pairs across 4 families	PASSED: tuned prior larger in 4 of 4
Our own $986\times$ headline	Is it typical of tuned models, or the extreme?	KILLED its typicality: the band is 19–$986\times$
The "undetermined answer" reading	Restrict to models whose canary is clean, as the prereg demands	PASSED: base 3.3–$3.6\times$, tuned 23.7–$986.5\times$, no overlap
The enumerate arm itself	Re-run Qwen2.5-3B and diff against the committed artifact	PASSED: worst $\lvert\Delta\rvert$ 0.000000
"Sample a few orderings instead"	Subsample the enumerated population, 4000 draws per k	KILLED it: k=4 recovers 4% of the spread
Our own Latin-square fix	Does it beat a random draw of the same size?	KILLED it: 8 of 16, a tie
The introspection probe	Ask the identical question about the PHASE OF THE MOON	KILLED 8 of 16 checkpoints
The introspection probe	Reverse the wording: belief, or agreement?	KILLED 3 more
The $\rho = -0.714$ introspection result	Exact permutation test, all 720 relabellings	KILLED its significance: $p = 0.1361$
The erase result	`erase_only`: does projecting a direction out move the probe on its own?	KILLED 2 of 4 erase layers
The erase profile	Ratio of primary to erase-artifact across layers	PASSED, post hoc
Our own erase operation	Does removing one direction actually remove the property?	KILLED the word "erase": refit cv 1.000 at every k
The subspace-erase result	A RANDOM subspace of the same rank	PASSED: random removes ≈ 0, fitted removes 69%
Every arm's null	Capability gate: can the readout move at all?	KILLED two arms outright
Every arm's statistic	Planted effect of known size at full strength and at the detection floor	PASSED
The analysis pipeline	Shuffle condition labels, rerun the analyzer	PASSED: returns no verdict

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Result under test	Control (what it could have shown)	Verdict
This paper's own number-checker	Break each checked number in a copy of `main.tex`: does the checker notice?	PASSED: fires on all 8, and on the untouched paper does not
The negative-pole null at 0.67 depth	Eight-depth sweep against Venkatesh [2026]	PASSED, then moot after replication
Arm B (prefilled continuation)	Capability criterion over five stems	KILLED the arm: gate 0.00000
Llama as an evaluation model	Peak mean pole shift against a 0.02 bar	KILLED the arm: inert

C Reproducibility

All fourteen preregistrations, every raw artifact committed unscored, every analyzer, and 339 tests are at `https://github.com/collapseindex/report-gap`. Each arm is one `modal_*.py` runner that computes nothing and one `analyze_*.py` scorer that holds every endpoint and gate; the split is preregistered, on the grounds that a runner which also decides can decide differently once it has seen the numbers. Runners stream to a volume and commit per batch, so an interrupt costs one batch. Every artifact records the package directory, a SHA-256 over the package source, and a scoped SHA-256 over the frozen stimuli that arm consumes.